Telecom Customer Churn Prediction

End-to-End Data Analytics Project — Problem Statement

1. Background

Customer churn — when a customer stops using a company's services — is one of the most critical challenges faced by telecom companies. Acquiring a new customer typically costs significantly more than retaining an existing one, which makes early identification of at-risk customers a high-value business problem. This project analyzes historical customer data from a telecom company to understand churn behavior and build a predictive model that can flag customers who are likely to leave.

2. Problem Statement

The telecom company has no systematic way of identifying which customers are at risk of churning before they actually leave. This project aims to analyze customer demographic, account, and service-usage data to uncover the key factors driving churn, and to build a classification model that predicts whether a customer is likely to churn — enabling the business to take proactive retention action.

3. Objectives

- Clean and prepare the raw customer dataset using Python (pandas) — fixing data types, handling missing values, and engineering new features (e.g. tenure groups, service usage count).
- Perform in-depth business analysis using SQL to answer key questions about churn drivers and customer segments.
- Design an interactive Power BI dashboard to visualize churn patterns and communicate insights to business stakeholders.
- Translate findings into actionable retention recommendations.

4. Dataset

The project uses the IBM Telco Customer Churn dataset (7,043 customers, 21 attributes), which includes customer demographics (gender, senior citizen status, partner/dependents), account information (tenure, contract type, payment method, billing), subscribed services (phone, internet, online security, tech support, streaming, etc.), and the churn label.

5. Key Business Questions (SQL Analysis)

- What is the total number of customers in the dataset?
- What percentage of customers have churned vs retained?
- Which contract type (month-to-month, one year, two year) has the highest churn rate?
- What is the average tenure of churned customers compared to retained customers?
- Which payment method has the highest churn rate?

- Do senior citizens churn more than non-senior citizens?
- Who are the top 10 customers by monthly charges?
- Which internet service type has the highest churn rate and average monthly charges?
- Do customers with zero additional services churn more than customers with multiple services?
- What is the churn rate across different tenure groups (0-12, 13-24, 25-48, 49-60, 60+ months)?
- Does paperless billing affect churn rate?
- What is the average monthly charge of churned vs retained customers?
- Which combination of contract type and internet service has the highest churn rate?
- What percentage of customers have multiple lines, and does that affect churn?
- Is there a difference in churn rate between customers with and without a partner/dependents?

6. Tools & Technology

Python (pandas) for data cleaning, handling missing values, data type correction, and feature engineering. **SQL** for structured business analysis and answering key churn-related questions. **Power BI** for building an interactive churn-insights dashboard.

7. Expected Outcome

A cleaned dataset ready for analysis, a set of clear business insights into churn drivers derived through SQL analysis, and an interactive Power BI dashboard that helps the business identify and prioritize at-risk customers for retention efforts.